

Chirag Venkatesh

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Analyst | Growth & Capital Development

Technical Skills

- **Analytical & Quantitative Capabilities:** Structured Data Analysis, Statistical Modeling, Performance Optimization, Quantitative Logic, Critical Trend Evaluation, Systemic Latency Minimization
- **Process Automation & Pipelines:** Workflow Optimization Scripting, Data Report Parsing, Log Validation, Automated Execution Builds
- **Languages & Frameworks:** Python, SQL (Concepts), C, MATLAB, React 19, Vite 7, Tailwind CSS
- **Core Framework Exposure:** Multi-Configuration Modeling (MCM), Operational Risk Mitigation, Cross-Functional System Integration, High-Speed Communication Protocol Basics

Professional Experience

External Consultant | Hardware Systems Operations Analyst

Dec 2024 – Aug 2025

Bangalore, India

- Conducted high-level performance and structural analysis on complex integrated systems, leveraging quantitative methodologies to drive processing and execution efficiency.
- Evaluated large-scale system diagnostic logs and validated design constraints to systematically resolve cross-functional performance failures.
- Built end-to-end automation scripts in Python and TCL to parse complex multi-source validation reports, reducing standard turnaround and optimization times.
- Collaborated across multi-configuration environments to guarantee deterministic behaviors and optimize system-level operational workflows.
- Supported product manufacturing workflows, validation procedures, and high-growth technology rollouts by ensuring asset consistency under varied stress parameters.
- Applied structured foundational principles of system design and digital logic to maximize communication integrity and scalability.

Bharat Earth Movers Ltd. (Rail & Metro)

Aug 2024 – Nov 2024

Software Engineer – Operations & Infrastructure Integration

Bangalore, India

- Worked on critical Train Control and Monitoring Systems (TCMS), managing cross-functional integration of distributed onboard subsystems to achieve real-time operational safety.
- Engineered and audited signaling modules over distributed CAN network nodes, establishing low-latency infrastructure and stable communication pathways.
- Partnered with multidisciplinary engineering teams to maintain system-level integration metrics, verifying alignment with rigorous architectural boundaries.
- Optimized network data workflows to support deterministic latency thresholds, directly enhancing operational scalability and data pipeline throughput.
- Directed systemic verification routines emphasizing extreme fault tolerance, mitigation of operational risk, and long-term asset stability under stress.

Projects

Sneaki – AI-Powered Productivity Intelligence Platform

- Architected an automated operations utility tracking raw application/window metrics and translating user activity into analytical event logs.
- Formulated data pipelines to process real-time webhook payloads, passing information to LLM backends to map operational categories and productivity metrics.
- Generated structured time-duration matrices and granular metadata event feeds to track resource allocation metrics.
- Implemented error-handling models and flexible hosting mechanics to ensure high operational availability and platform stability.

Statistical Arboreal System Pipeline

- Built an advanced image processing framework utilizing specialized mathematical descriptors for resilient pattern classification under environmental fluctuations.
- Engineered data comparative matrices to benchmark incoming visual parameters against classified reference datasets to compute precision mapping scores.
- Created a mathematical density estimator model to summarize aggregated signal distributions over sequential historical data arrays.
- Mapped continuous probability scores directly to dynamic operational control configurations to support responsive automated intervention loops.
- Authored formal analytical findings and empirical validation metrics in a peer-reviewed research conference publication.

Readimentary (RSVP Application Framework)

- Produced a web reader using custom rendering frameworks and velocity metrics to optimize cognitive focus and document parsing workflows.
- Engineered automated parsing modules and regional persistence architectures using browser database frameworks for long-term state integrity.
- Developed progress-tracking logic blocks, session markers, and sequential navigation routines to optimize state retention.
- Shipped a responsive, modern architecture utilizing React 19, Vite 7, and optimized rendering engines.

Real-Time Low-Latency Control Infrastructure

- Designed an integrated hardware-software ecosystem facilitating physical coordinate mapping with low-latency communication feedback.
- Deployed data smoothing routines and signal processing filters in Python to guarantee spatial stability and signal noise reduction.
- Integrated MQTT messaging infrastructure to synchronize distinct physical network configurations cleanly across disparate terminal endpoints.
- Calibrated response response curves to eliminate mechanical variance and establish high-fidelity positioning accuracy.

Education

PES University

B.Tech in Electronics and Communication Engineering | Minor in Computer Science Engineering

Bangalore, India

2020 – 2024